

Algebra 2: Unit 6 Review — ANSWER KEY

1. Solve: $x^{\frac{1}{3}} = 3$
 $(x^{\frac{1}{3}})^3 = 3^3$

Answer: $x = 27$

2. Solve: $(x + 1)^{\frac{1}{3}} - 2 = 3$
 $(x + 1)^{\frac{1}{3}} = 5$
 $x + 1 = 125$

Answer: $x = 124$

3. Solve: $2\sqrt[3]{x} + 1 = 5$
 $2\sqrt[3]{x} = 4 \rightarrow \sqrt[3]{x} = 2$
 $(\sqrt[3]{x})^3 = 2^3$

Answer: $x = 8$

4. Solve: $\sqrt[3]{2x + 5} = \sqrt[3]{4x - 9}$
 $2x + 5 = 4x - 9$
 $14 = 2x$

Answer: $x = 7$

5. Solve: $(2x - 14)^{\frac{1}{3}} = (x + 11)^{\frac{1}{3}}$
 $2x - 14 = x + 11$

Answer: $x = 25$

6. Solve: $3(x - 2)^{\frac{1}{3}} = 6$
 $(x - 2)^{\frac{1}{3}} = 2$
 $x - 2 = 8$

Answer: $x = 10$

7. Solve: $5 - x^{\frac{1}{3}} = 8$
 $-x^{\frac{1}{3}} = 3 \rightarrow x^{\frac{1}{3}} = -3$
 $x = (-3)^3$

Answer: $x = -27$

8. Solve: $\sqrt{x} = -3$
The principal square root cannot be negative.
Answer: No solution

9. Solve: $3(7x + 1)^{\frac{1}{3}} = 4(3x)^{\frac{1}{3}}$
 $27(7x + 1) = 64(3x)$
 $189x + 27 = 192x$
 $27 = 3x$

Answer: $x = 9$

10. Solve: $(x^2 - 3x - 10)^{\frac{1}{3}} = 2$
 $x^2 - 3x - 10 = 8$
 $x^2 - 3x - 18 = 0$
 $(x - 6)(x + 3) = 0$

Answer: $x = 6, -3$

11. Solve: $\sqrt[3]{x^2 - 2x - 7} = 1$
 $x^2 - 2x - 7 = 1$
 $x^2 - 2x - 8 = 0$
 $(x - 4)(x + 2) = 0$

Answer: $x = 4, -2$

12. Solve: $(2x - 1)^{\frac{1}{3}} = 5$
 $2x - 1 = 125$
 $2x = 126$

Answer: $x = 63$

13. Solve: $\sqrt[3]{x^2 - 2x} = -1$
 $x^2 - 2x = (-1)^3 \rightarrow x^2 - 2x = -1$
 $x^2 - 2x + 1 = 0 \rightarrow (x - 1)^2 = 0$

Answer: $x = 1$

14. Solve: $10 = 2\sqrt[3]{h - 25}$
 $5 = \sqrt[3]{h - 25}$
 $125 = h - 25$

Answer: $h = 150$

15. Solve: $2\sqrt[3]{5x + 12} = 3\sqrt[3]{2x + 2}$
 $8(5x + 12) = 27(2x + 2)$
 $40x + 96 = 54x + 54$
 $42 = 14x$

Answer: $x = 3$